9.26

Exp 12(b)

Chat client - Server using TCP Sockets

Aim:

To Implement a simple chat client - Server application using TCP sockets

Algorithm:

Server:

1. create a TCP socket.
2. bind and listent for client
3. Accept the connection
4. Recieve a Message from client
5. send a reply to the client

Client:

1. Create a Tcp socket
2. Connect to the Server
3. Send a Message
4. Recieve the server's reply
5. Display the reply
6. Close the connection

Input:

Client: Hello
Server: Hi
Client: How are you?
Server: Fine
Client: bye

Output

Client: Hello
Server: Hi
Client: How are you?
Server: Fine

Code - Server

```
Import Socket
S = Socket.socket()
S.bind("localhost", 5000)
S.Lisent(1)
C.addr = S.accept()
While true:
    Msg = C.recv(1024).decode
    if msg = "bye": break
    Print("Client", Msg)
C.close()
S.close()
```

Code - Client

```
Import Socket
S = Socket.socket()
S.connect("localhost" 5000)
While true:
    Msg = input("Client:")
    S.Send(Msg.encode())
    if Msg = "large":
        break;
S.close()
```

Result:

Thus the chat client server was successfully implemented using TCP socket.